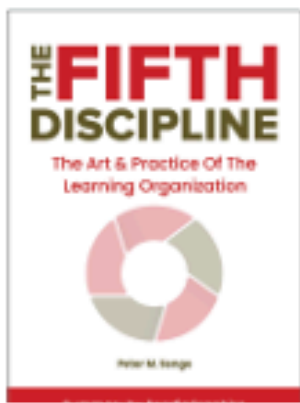




THE FIFTH DISCIPLINE

The Art & Practice of
The Learning Organization

Peter M. Senge

The Big “So What”



In today's increasingly dynamic and complex world, an organization's ability to learn and renew itself is a sustainable competitive advantage. This book provides the blueprints for the “learning organization”, which is a potentially better model for:

- Managing and leading **change**
- Building **adaptive organizations** that can cope with an increasingly connected and volatile world
- Improving **performance and happiness** in the workforce

Introduction



A learning organization is “where people continually expand their capacity to create the results they truly desire, where new and expansive patterns of thinking are nurtured, where collective aspiration is set free, and where people are continually learning how to learn together.”



Learning organizations combine adaptive learning (that help us respond and adapt to environmental changes) with generative learning (that enhances our capacity to create). In so doing, **they continually expand their capacity to create their future.**

As the world becomes more dynamic and connected, organizations that thrive and excel are those that tap on people's ability and commitment to learn, at all levels. This is highly achievable as we are all learners by nature.

KEY QUOTES

“In the long run, the only sustainable source of competitive edge is your organisation's ability to learn faster than its competitors.”

“Through learning we extend our capacity to create, to be part of the generative process of life. There is within each of us a deep hunger for this type of learning.”

“There are ways of working together that are vastly more satisfying and more productive than the prevailing system of management.”

The 5 Learning Disciplines and 7 Learning Disabilities

The 5 Disciplines of the Learning Organization

There are 5 basic disciplines that set learning organizations apart from “traditional organizations”. These are elaborated in the subsequent pages.

1. **Systems Thinking**, i.e. seeing underlying systemic structures. This is the “Fifth Discipline” that integrates all the disciplines.
2. **Personal Mastery**: Continually learn, achieve clarity & depth of vision, see reality objectively, & close the reality-vision gap
3. **Mental Models**: Uncover limiting beliefs & flaws in our world view
4. **Shared vision**: Commitment to a shared long-term aspiration
5. **Team learning**: Align & develop the capacity of the team as a whole, building on individual talent & vision



In applying the 5 disciplines, it is important to understand: A discipline is a *path* for developing skills or competencies, and practicing it is a **lifelong process**.



These disciplines are **personal disciplines** that **determine how we think, interact and learn with one another**. Our personal disciplines combine to define the organizational disciplines.

The 7 key learning disabilities



“I am my position”: Most people focus on their job title or daily tasks, not the purpose of their enterprise. When people focus on their positions, they don’t see their impact on the whole and feel little responsibility for organizational results.



“The enemy is out there”: When we focus only on our positions, we don’t see how our actions generate consequences that boomerang back to hit us. Instead, we blame others.



The illusion of taking charge: When we take aggressive action to fight the “enemy out there”, we are actually being reactive. True proactiveness starts in our way of thinking.

KEY QUOTES

“In building learning organizations there is no ultimate destination or end state, only a lifelong journey.”

“Organizations work the way they do because of how we work, how we think and interact; the changes required ahead are not only in our organizations but in ourselves as well.”



The fixation on events: Monthly sales, quarterly earnings etc. are all short-term events that **distract us from the larger, longer-term patterns and causes that lie behind those events**. If we focus on events, we can only predict and react to them; we cannot create.



The parable of the boiled frog: A frog in a pot of water that is slowly brought to boiling point will not try to escape. Likewise, we won't notice the **slow, gradual changes around us (which often pose the greatest threats) unless we slow down to observe them**.



The delusion of learning from experience: We learn best from direct experience. The problem is, **we usually don't directly experience (and learn from) the consequences of important organizational decisions (e.g. R&D, leadership changes) because they have a complex ripple effect over long periods of time**.



The myth of the management team: The organization is supposedly led by capable management with diverse expertise. However, most of us are trained *not* to admit that we don't know the answer. The reality is, **we often have management teams who are pretending to know, rather than actively learning**.



Discipline 1: Systems Thinking

This is the "Fifth Discipline" or **the cornerstone of the learning organization** that underlies and unifies all 5 principles. Our education (and eventually management) systems have trained us to break down problems into small parts. This makes complex tasks manageable, but we no longer see connections and the big picture.

Systems influence behaviour. To understand a thunderstorm, we need to understand the entire eco-system, not just part of the pattern. **When faced with a problem, we also need to understand the true systemic causes, not just "solutions" close by or visible to us**. There are 3 progressive levels of thinking: Events (reactive) => Patterns of Behaviour (responsive) => Systematic Structure (generative).

KEY QUOTES

"All too often, proactiveness is reactivity in disguise... True proactiveness comes from seeing how we contribute to our own problems."

"We would never transform the prevailing system of management without transforming our prevailing system of education. They are the same system."
- Dr. W. Edwards Deming

"When placed in the same system, people, however different, tend to produce similar results."

A Shift of Mind

Using systems thinking as a thinking cornerstone brings a **shift from seeing parts to wholes, from reacting to creating**. It means organizing multiple variables and details into a coherent story, where problem sources are identified and can be properly remedied. It means **concurrently seeing the trees and the forest, the detailed and broad patterns**.

We tend to see reality in straight lines, when it is really made up of curves and circles. Language shapes perception. When we get fluent in the language of systems thinking, **we start to see circles of causality**:



We see **inter-relationships** rather than linear cause-effect chains



We see **processes of change** rather than static snapshots

The main shift is to understand what we are **part of the feedback process, not outside of it**.

Building Blocks of "System Archetypes"

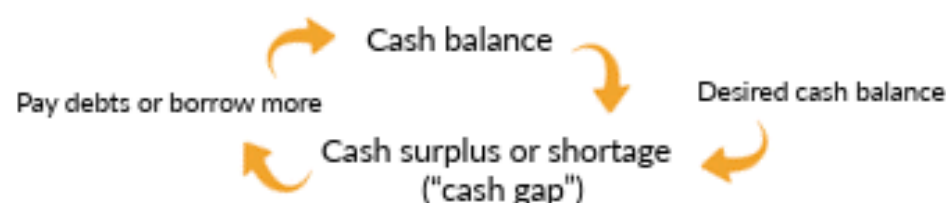
To practice systems thinking, **we must learn to see recurring structures or system archetypes**. Variables (including our actions) can be organized in loops of cause-effect relationships, where they reinforce or balance one another through a "feedback" process. There are **3 key components that form system archetypes**:



Reinforcing (or amplifying) feedback: These are engines of growth that amplify small changes/ actions to form vicious or virtuous cycles e.g.



Balancing (or stabilizing) feedback: These are part of a self-correction process to maintain an implicit/explicit goal/target. They are sources of both stability and resistance e.g.



Both explicit and implicit balancing processes are constantly at play. Most resistance comes from implicit goals/assumptions (e.g. ambitions, fears).

KEY QUOTES

"(Systems thinking) means organizing detail complexity into a coherent story that illuminates the causes of problems and how they can be remedied in enduring ways."

"The practice of systems thinking...recognize types of 'structures' that recur again and again."



Delays: There may be systemic delays or “time lags” between actions and visible results (e.g. beer cannot be brewed instantaneously). A lack of awareness may trigger overly-aggressive actions that cause us to “overshoot” our mark.

Generative Learning through Systems Thinking



We cannot address issues that we don't see. Once we see the full structures within which we operate, we can work with or change them, slowly freeing ourselves from previously-unseen forces.

There are only a relatively small number of systems archetypes that underlie most management problems. With practice, we can learn to quickly recognize these archetypes, and know the high and low points of leverage that we can use.

Appendix 2 in the book lays out a list of common system archetypes and the management principles for each situation. Here are 2 of the most common archetypes:

- **Limits to Growth.** Often, we see periods of growth then a slow down, halt or reversal, as the growth hits certain limits (e.g. lack of manpower/supporting processes). The solution is not to push growth, but to identify and remove the factors limiting growth.
- **Shifting the Burden.** It is usually hard to identify the true underlying problem(s), or costly to address them. Hence, managers may prefer to “shift the burden” using symptomatic solutions. These may seem to alleviate the issues, but in reality only move the problems to other parts of the system. The underlying problems usually resurface in a different or worse form.

Using the archetypes, identify strong leverage, where small but well-focused actions can bring huge and enduring improvements.



Discipline 2: Personal Mastery

Learning organizations can only exist through individuals who learn and exhibit high levels of personal mastery, i.e. they live life creatively (rather than reactively), and keep recreating themselves through improvement in competence, skills, spiritual growth etc. Their spirit for learning and growth form that of the organization's. Pursuing personal mastery is a lifelong process...you never “finish” learning.

KEY QUOTES

“Structures of which we are unaware hold us prisoner.”

“Being stuck in a system that is not working invariably leads to (us) feeling frustrated and trapped – until we see the larger patterns and our own part in creating these patterns. Once we do, new alternatives become evident.”

“To change the behaviour of the system, you must identify and change the limiting factor.”

“Individual learning does not guarantee organizational learning. But without it no organizational learning occurs.”

Elements of Personal Mastery



Personal Vision: This is a specific picture of your desired future, including areas like health, freedom etc. It includes (but is more than) your life purpose, and has a deep meaning for you.



Holding creative tension: "Creative tension" is the gap between your vision and reality. To resolve this tension, you can either work to bring your reality up towards the vision, or compromise and bring your vision down towards reality. Mastering creative tension means gaining clarity on your vision and your reality, using the gap to generate energy for change, and developing a capacity for perseverance and patience.



Overcoming structural Conflict: Most of us have at least 1 of these beliefs limiting our ability to create what we really want:

- Powerlessness: That we are unable to create the things we want
- Unworthiness: That we do not deserve to get what we want

You can overcome such resistance with strong willpower, though such limiting beliefs can also be changed fundamentally with time.



Commitment to the truth: The best way to start dealing with structural conflict is to be willing to see how we limit or deceive ourselves, to continually challenge our theories of why things are the way they are, and to broaden/ deepen our awareness of structures underlying current events.



Using the subconscious: Deliberately develop rapport with your subconscious or automatic minds, by focusing on your desired results clearly, in a quiet state of mind.

Fostering Personal Mastery in an Organization

You can't force someone to embark on the path of personal growth. As a leader, you can encourage personal mastery by:



Creating a conducive climate e.g. make it safe for people to create visions, inquire & commit to the truth, challenge status quo etc.



Developing all 5 learning disciplines concurrently in the organization, so they can strengthen/ reinforce one another.



Being a role model by committing to your own personal mastery.

KEY QUOTES

"People with high levels of personal mastery are continually expanding their ability to create the results in the life they truly seek. From their quest for continual learning comes the spirit of the learning organization."

"Vision is a specific destination, a picture of the desired future... it is the goal pulling you forward that makes all the work worthwhile."

"Personal mastery...means approaching one's life as a creative work, living life from a creative as opposed to reactive viewpoint."

"There's nothing more powerful you can do to encourage others in their quest for personal mastery than to be serious in your own quest."



Discipline 3: Mental Models

We simplify our world and carry it in our heads as images, assumptions and stories e.g. "People are lazy by nature". **Mental models shape how we act because they affect what we see.** Two people with different mental models can experience the same event, but see different details and make different interpretations.

There are no right or wrong mental models - the problems arise when they become implicit and we are unaware that they are shaping our thinking and actions. **What we are unaware of, we cannot examine, and hence cannot change or improve.** The discipline starts with turning the mirror inward to discover our own mental models.



Surfacing and Testing Mental Models

There are 3 facets to develop an organization's capacity to surface and test mental models:

- **Use tools & skills** that promote personal awareness & reflective skills. Specifically, **reflective skills** involve becoming more aware of how we form our mental models & how they influence our actions. **Inquiry skills** involve how we interact with others, in dealing with complex and conflictual issues. Some key tools include:



Reflect on gaps between espoused theory (what we say) vs theory-in-use (what we do): When we identify a gap or incongruence, don't judge or criticise. Instead, ask: "Do I really value the espoused theory and is it really a part of my vision?", and make adjustments from there.



Recognize "leaps of abstraction", i.e. our jumps from concrete observations to generalization. Such untested generalizations (e.g. "customers don't care about speed") can easily lead to further generalizations. To spot leaps of abstraction, ask yourself:

1. What do I believe about the way the world works, and upon what "data" is this generalization based on?
2. Am I willing to consider that this generalization may be inaccurate or misleading?

KEY QUOTES

"Structures of which we are unaware hold us prisoner. Once we can see them and name them, they no longer have the same hold on us. This is as much true for individuals as it is for organizations."

*"Our theories determine what we measure."
- Albert Einstein*

"Generative learning...requires people at all levels who can surface and challenge their mental models before external circumstances compel them to do so."

"Because we remain unaware of our mental models, the models remain unexamined. Because they are unexamined, the models remain unchanged. As the world changes, the gap widens between our mental models and reality, leading to increasingly counter productive actions."



Expose the “left-hand column” (express what we normally don’t say): Record a conversation by writing down what’s being said on the right hand column. Then, write down what you were thinking/feeling (that led to your spoken response) on the left hand column. Once you can see your assumptions (and why you said or chose to say/not say certain things), you can choose to more openly articulate your views.



Balancing inquiry and advocacy. For effective collaborative learning, we need both advocacy (taking a stand or a point of view) and inquiry (asking questions and inviting alternative views). An example of a balanced approach is: “This is my point of view and how I arrived at it. How does it sound to you?”

- **Develop “infrastructures”** that institutionalize regular use of tools and skills for the discipline of mental models, and
- **Build a culture** that promotes inquiry & challenges our thinking



Discipline 4: Shared Vision

A vision is truly shared when each person in the organization has a similar picture, cares deeply about it because it reflects their personal vision, and is connected to others by this common aspiration.

Shared vision is crucial for the learning organization because it provides the energy and focus for learning, and links people’s work to a larger purpose embodied in the organization’s products and services. It creates alignment between people and organization, creates trust and courage to do what is needed for the vision, and fosters a naturally long-term view in people.

Building Shared Vision



- **Encouraging personal vision:** Shared visions derive their energy and commitment from personal visions, and the relentless pursuit of the vision and growth that comes with personal mastery.
- **Personal vision => Shared vision:** The relationship between personal vision and shared vision is best illustrated using a hologram. When you divide a hologram, each small piece still shows the whole image intact. **When people share a vision, each of them shares responsibility for the whole vision** (not just for his

KEY QUOTES

“Practicing inquiry and advocacy means being willing to expose the limitations in your own thinking – the willingness to be wrong. Nothing else will make it safe for others to do likewise.”

“Few, if any, forces in human affairs are as powerful as shared vision.”

“A shared vision is a vision that many people are truly committed to, because it reflects their own personal vision.”

“A vision is truly shared when you and I have a similar picture...when people truly share a vision they are connected, bound together by a common aspiration.”

“Visions are exhilarating. They create the spark, the excitement that lifts an organization out of the mundane.”

piece), each represents the image from a different point of view, enriching the vision without changing it. Visions that are truly shared don't emerge overnight; they grow out of expression, listening and interaction of individual visions.

- **Spreading visions: commitment vs compliance.** People who are genuinely compliant accept the vision and play by the rules of the game. However, a committed person takes it further – he truly wants the vision, brings additional energy, passion and excitement, and takes responsible for the game. In fact, he will try to change the rules of the game, if necessary, to achieve the vision.

Getting Enrolment and Commitment to the Vision



As a leader, you can use these guidelines:

- Be enrolled yourself
- Describe the vision simply & honestly
- Let the other person choose

The vision cannot stand alone. It works together with an organization's purpose and core values to answer the question: What do we believe in? This is then translated into concrete day-to-day behaviours.

- Vision = "What" is the picture of the future we seek to create
- Purpose = "Why" our organization exists
- Core Values = "How" the company and its people lives on a day-to-day basis while pursuing the vision



Discipline 5: Team Learning



Alignment is a pre-requisite to individual empowerment. Empowering individuals when there is no or low alignment only brings chaos and wastes energy. Team learning aligns and develops the capacity of the team as a whole, building on individual talent and vision to achieve results that members can't achieve on their own.

In organizations, team learning involves 3 key dimensions:



Insightful thinking about complex issues by tapping the minds of many people.

KEY QUOTES

"This is the art of visionary leadership – how shared visions are built from personal visions."

"People who are enrolled or committed truly want the vision. Genuinely compliant people accept the vision."

"A group of people truly committed to a common vision is an awesome force"

"There is really nothing you can do to get another person to enrol or commit. Enrolment and commitment require freedom of choice."

"When a team becomes more aligned, a commonality of direction emerges. There is less wasted energy. In fact, a resonance or synergy develops."



Innovative and coordinated action, with operational trust in how members can depend on one another to complement their actions



Learning across teams. Team members usually interact and work with other teams too, hence spreading the learning across teams

The Discipline of Team Learning

Team learning involves :



Dialogue and discussions: Dialogue lets a group explore complex or difficult issues from many points of view, hence going beyond individual understanding and uncovering insight that isn't achievable individually. For a dialogue to work, all parties must:

- **Suspend assumptions**: be aware of assumptions and hold them up for examination
- **See each other as colleagues** (putting aside hierarchy and roles)
- **Balance dialogue & discussion**: Dialogues are diverging, with different views being presented and complex issues explored. Discussions are converging, with views being defended, analysis and decisions made. Learning teams switch between dialogue and discussion to identify and decide on new courses of action.



Dealing with conflict and defensiveness. Great teams do not avoid conflict. Instead, they develop skills to see and deal with current reality, and overcome defensiveness. That is why the disciplines of personal mastery and mental models are essential at an individual level before they can be amplified and leveraged at a team level.



Learning through practice. Learning teams need to learn how to learn together. Like team sports, they need "practice fields" to become more effective learning units, to form team skills and team IQ that exceed individual skills and IQ.

INTEGRATION AND APPLICATION

Integrating the 5 Disciplines

Organizations are living systems; they are not static – they have huge capacities to learn, evolve and heal themselves, by how its people live and work day to day.

KEY QUOTES

"The purpose of dialogue is to go beyond any one individual's understanding."

"Unless the team also builds the skills for seeing rather than obscuring current reality, their capacity for learning will be unreliable."

"Perhaps when we rediscover organizations as living systems, we will also rediscover what it actually means to us as human beings to work together for a purpose that really matters."

The 5 disciplines reinforce and support one another, integrated by **Systems Thinking**. As people practice the discipline of **personal mastery**, they experience gradual changes, e.g. integration of reason and intuition, greater connectedness to the world, greater compassion and commitment to the whole. The discipline of **mental models** help people to examine their own assumptions, become more open to and identify new ways of thinking. **Shared vision** helps people see how their actions contribute to changing and shaping their future. All 3 disciplines set the foundation for **team learning**, which helps team members to create results they desire, at a level beyond their individual capability. **Systems thinking** underlies all 4 other disciplines to help us see the big picture and our roles in it, restructure assumptions, and reveal causes and leverage in complex situations.

STRATEGIES

You can build deep learning cycles to reinforce current or emerging learning cultures. There are also many possible intervention strategies, but they should include the 3 elements of the strategic architecture to be most effective:



Strategies for building the learning organization

Integrate learning & working: Learning as an “add-on” to people’s existing work only creates more fragmentation. Instead, (a) create **opportunities to apply the tools to daily work** and (b) create an **environment that is conducive** to reflecting, thinking and building shared visions. Simple tools like After-Action-Reviews can help to connect action and reflection.

Start where you are, with whoever is there: Strategic thinking need not begin at the top. **Any leader, at any level, can/should think strategically.** You can start by **focusing on the “impossibles”**, i.e. problems that people have given up on. First, take time to **figure it out**, then **enrol people** who are willing to work with you voluntarily, and gradually build from there. To succeed, you’ll have to believe in and focus on **tapping people’s talents and deepest aspirations.**

KEY QUOTES

“Successful learning practitioners intent upon having a large-scale impact learn how to connect with the core of the organization –at the deepest levels of individual and collective identity–and how the organization most naturally creates value.”



Become Bicultural: Stay in touch with the wider organizational environment, as you need to **work with the political forces in the organization**. Start with **stealth transformation**. There's no need to hide your efforts, but don't make a big deal at the early stages until you have enough results to garner support. Be sensitive to and **use the language of the incumbent**, so you relate to deep values that people can understand.



Create Practice Fields, where people can try something new and make mistakes before deploying the ideas in the real-world. Examples include **"camp and hierarchy"** ("camps" where people play and address issues away from formal management systems), and **"the swirl and wall of fire"** (let ideas get tossed around until they get enough organization-wide support).



Connect with the Core of the Business, i.e. where the organization creates value most naturally. First, you must identify **who the organization is at the core** and what it stands for. Then, **do the "fieldwork"** to identify the **emerging leaders**/people in the organization who want to step up and run with the new ideas.



Build learning communities, i.e. groups who gravitate together to address questions they care about. These groups can't be controlled nor manipulated. Use **conversations with heart and meaning**, and a mix of dialogue and implementation.



Embrace diversity. Work with people/ organizations who are very different, to see varying perspectives and continually challenge our thinking. Consider **connecting to new partners across sectors**.



Develop Learning Infrastructures (e.g. training and formal education, practice, research and doctrine) that embed learning & insights deep into the fabric of the organization.

The Leader's New Work

Leaders in learning organizations are fundamentally designers, teachers and stewards.



Leaders as Designers: A leader is often considered the captain or navigator of a ship. However, **the most impactful and oft-neglected role of the leader is as the designer of the**

KEY QUOTES

"The new view of leadership in learning organizations centers on subtler and more important tasks. In a learning organization, leaders are designers, teachers and stewards."

ship. Anyone from any position can be a leader. Appreciating the organization as a living system, a good leader designs the ship with:

- **A strong foundation of governing ideas** (vision, purpose and values) that connect, mobilize and focus its people.
- **Iterative design and learning infrastructures** that evolve over time, (e.g. policies, strategies, structures that translate the ideas into daily decisions)
- **IT infrastructures** that are designed by the teams responsible for learning implementation, not by technical designers focusing on the technology itself.



Leaders as Teachers: As teachers, leaders help people achieve more accurate, insightful and empowering views of reality. To do so, they:

- **See gaps in organizational capacity**, through engaging personally in dialogues
- **Balance short and long term solutions:** They help people to apply systems thinking to solve complex problems, while recognizing that such solutions take time and resources. They tackle this dilemma by responding to short-term opportunities in ways that concurrently build longer-term capacity.
- **Become learners first.** They inspire their students through their commitment as practitioners, not as preachers or “advocates”.



Leaders as Stewards: Stewardship is about the desire to serve those we lead, as well as serve a larger purpose. Leaders who demonstrate stewardship are:

- **Committed without blind certainty:** Someone who is **certain** about his purpose may be fanatical or close-minded; someone who is **committed** to his vision recognizes his uncertainties but chooses to stay committed to his path.
- **Focus on the larger goal:** Learning-oriented leaders don't make decisions to preserve their own power, ambition/status. They focus on the longer-term results, depends on their clarity, commitment & openness to win over their followers

Other Useful Details in the Book to Look Out For



The book was first written in 1990. What seemed radical then is now increasingly accepted and practiced. This revised edition was written some 16 years later, with updates based on global changes and insights from practical application.

This summary only outlines the key ideas of the 423-page book. There are many other details included in the book, such as:

KEY QUOTES

“It is fruitless to be the leader in an organization that is poorly designed.”

“(Artful designers) see visions and other guiding ideas as tools for mobilizing and focusing energy. They judge the ideas by their impact, not by how they sound. And they never forget that guiding ideas are always works in progress.”

“Genuine commitment...always co-exists with some element of questioning and uncertainty. In this sense, commitment is truly a choice rather than a compulsion.”

- 11 Laws of the Fifth Discipline
- Details of “The Beer Game” (a laboratory experiment that shows the learning disabilities at work in real organizations and systems)
- Real-life case studies to illustrate the 5 disciplines at work
- Snippets and takeaways from Senge’s conversations with leaders and organizational learning practitioners.
- How the fifth discipline can be used to address many of the social and environmental problems that we face today, from climate changes to entrenched poverty cycles.

About the Author



Peter Michael Senge (born 1947) is an American systems scientist. He is also the founding chairperson of the Society for Organizational Learning, a senior lecturer at the MIT Sloan School of Management, and co-faculty at the New England Complex Systems Institute.

Senge lives in Cambridge, Massachusetts. Besides authoring the book *The Fifth Discipline: The Art and Practice of the Learning Organization* (1990, rev. 2006), he also co-authored *The Fifth Discipline Fieldbook*, *The Dance of Change*, and *Schools That Learn*.

KEY QUOTES